

# Raj Modi

Full-Stack Software Developer | TypeScript · React/Next.js · FastAPI · PostgreSQL · DevOps & AI

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## SUMMARY

Final-semester B.Tech (CSE) student who **designs, develops, and deploys full-stack web-app MVPs** end-to-end — proficient in **TypeScript** and **database technologies** (PostgreSQL, MySQL), with hands-on **DevOps** (Docker, CI/CD, DigitalOcean) and **AI** (RAG, semantic matching, MCP). Experienced **managing user databases** and **integrating existing back-ends** into cloud platforms. Fast, curious learner who ships working software over theory — a strong fit for AEC-sector cloud development.

## TECHNICAL SKILLS

**Languages:** TypeScript, JavaScript (ES6+), Python, SQL, Java  
**Front-End:** React, Next.js, Vue.js, Tailwind CSS, Ant Design, shadcn/ui, Zustand, Recharts / ECharts, Three.js  
**Back-End:** Next.js API Routes, tRPC, FastAPI, Node.js, REST API design & integration, WebSockets, NextAuth / JWT Auth  
**Database & ORM:** PostgreSQL, MySQL, DuckDB, Redis; Prisma, Drizzle ORM  
**DevOps & Cloud:** Docker, Docker Compose, DigitalOcean, Vercel, GitHub Actions (CI/CD), Git/GitHub  
**AI / ML:** RAG, Semantic Matching, MCP (Model Context Protocol), LangGraph Multi-Agent, XGBoost / SHAP, spaCy NLP

## PROJECTS

### Keystone – AEC Project Platform with Conflict-Aware Document AI

[Live Demo](#) | 2026

**Multi-Tenant Full-Stack SaaS** | **Novel Conflict-Aware RAG** | Next.js 14, TypeScript, tRPC, Prisma, PostgreSQL/pgvector, shadcn/ui, Tailwind CSS, Docker, Vercel

- Designed, developed, and deployed a **multi-tenant cloud-platform MVP** for Architecture/Engineering/Construction firms — **end-to-end TypeScript** (Next.js 14 App Router + tRPC + Prisma + PostgreSQL) to manage projects, drawings/specs, and RFI workflows, with org-scoped **role-based access** (Architect/Engineer/Contractor/Client) over a managed **users database**.
- Built a novel **conflict-aware RAG** engine that clusters retrieved evidence to **surface contradictions between specs and drawings** (vs. a single confident answer), plus a **change-impact graph** computing the downstream “blast radius” of any document revision — via **pgvector semantic search**, an **MCP server**, and source-cited answers; **deployed live on Vercel + Neon** (GitHub Actions CI/CD) with a **shadcn/ui + Tailwind** UI.

### Verita – Financial Crime & Compliance Intelligence

2026

**Full-Stack TypeScript + ML + CI/CD** | React, TypeScript, FastAPI, PostgreSQL, DuckDB, XGBoost, Docker

- Built and deployed a full-stack analytics platform (**React + TypeScript**, FastAPI, **PostgreSQL + DuckDB**) with real ML fraud scoring (**XGBoost**, ROC-AUC ~0.99), SHAP explainability, and natural-language→SQL — **Dockerized** with GitHub Actions **CI** (82 tests passing).

### AgentForge – Multi-Agent AI Planning Engine

2026

**TypeScript Front-End, Users DB & Multi-Agent AI** | React + TypeScript, FastAPI, LangGraph, PostgreSQL, Redis, Docker Compose, JWT, MCP

- Engineered a 7-agent **LangGraph** system with a **React + TypeScript** front-end and real-time WebSocket streaming; **containerized the full stack** (FastAPI, **PostgreSQL**, Redis) for **one-command Docker Compose** deployment.
- Implemented **JWT authentication** and a managed **users database** with protected, role-based routes, plus a custom **MCP server** exposing agent tools.

### MissionOS – Systems Engineering & Project Lifecycle Platform

2026

**Multi-Project Lifecycle Management Web App** | JavaScript (SPA), Vite, Canvas, Chart.js, Selenium

- Built a single-page web platform managing the full **engineering project lifecycle** across 6 concurrent programs — tracking **247 requirements**, 8 interface-mapped subsystems, a 5×5 risk matrix, and multi-criteria trade studies through 11 lifecycle phases, with authenticated access and a KPI command-deck dashboard.
- Hardened with a **58-test Selenium suite** and built as a modular, dependency-mapped architecture.

### Nyay AI – Unified Legal Intelligence Platform

2026

**Hybrid RAG & Semantic Search** | Python, Flask, Hybrid RAG, Knowledge Graph, LLaMA 3.3

- Built a **hybrid RAG** pipeline — dense-vector + BM25 **semantic matching** with reciprocal-rank fusion, HyDE query expansion, and CRAG self-correction — over legal documents in 22+ Indian languages, returning source-cited, streaming answers.

## WORK EXPERIENCE

### Software Developer & Project Management Lead

Jun 2025 – Feb 2026

VW Group Academy (Škoda / Volkswagen)

Remote, India

- Led a **3-student team** as project-management lead and key developer on an 8-month engagement with a **real enterprise client**: gathered requirements and built an automated scheduling engine (ETL + constraint-based allocator) for **~1,243 annual training events** across 3 centres — reaching **98.5% auto-allocation** and cutting manual effort **~80%**.

### Community & Growth Intern

2023

Engineer's Cradle Pvt. Ltd.

Remote, India

- Lifted user engagement **40%** via data-driven campaigns and automated reporting.

## ACHIEVEMENTS & CERTIFICATIONS

- SIH 2025 – Rank 63 (Top 10%)**, Smart India Hackathon (accessible Kolam designer with voice assistant) | **AIR 8**, National Creative Olympiad.
- Certifications:** Google – Advanced Data Analytics | IBM – Data Visualization with Python | Indiana University – Financial Planning.
- Joint President**, Student Alumni Relations Society, MIT-WPU (2025–26) | **Execution Head**, RIDE 2025 tech fest (500+ attendees).